

Race

2026-06-12

Packages

```
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.3
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(tidyr)  
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.5.3
```

```
library(DataExplorer)
```

```
## Warning: package 'DataExplorer' was built under R version 4.5.3
```

```
library(knitr)  
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.3
```

```
##  
## Attaching package: 'kableExtra'
```

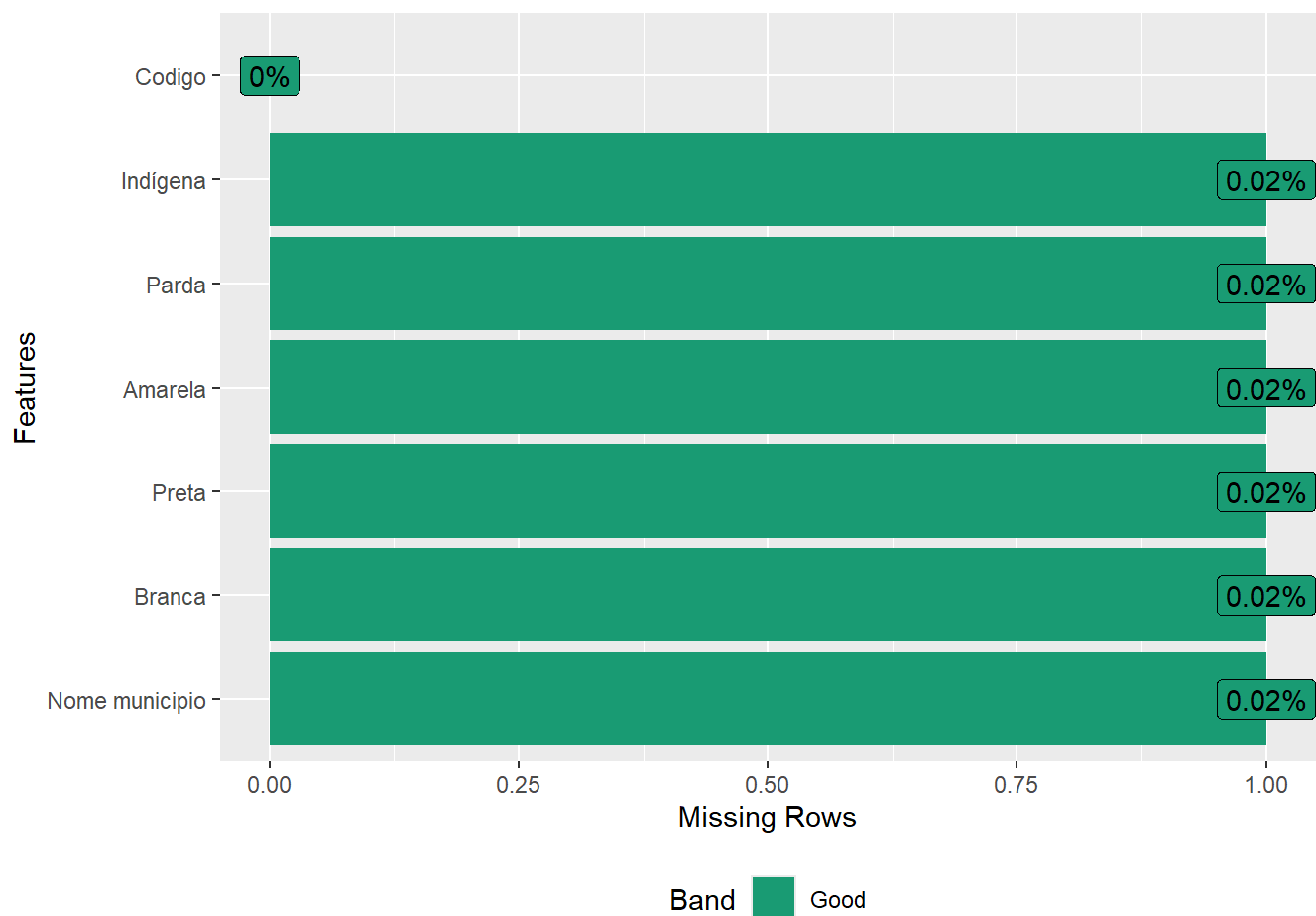
```
## The following object is masked from 'package:dplyr':  
##  
##      group_rows
```

Race Dataset

```
race <- read_excel("C:/Users/alyss/Desktop/Brasil/Analysis/dados_current - 2026-06-09/pop_raca_c  
or_2022_clean.xlsx")
```

Clean

```
plot_missing(race)
```



```

race_sp <- race %>%
  filter(Codigo >= 350000 & Codigo < 360000)

library(dplyr)

race_sp <- race_sp %>%
  mutate(
    across(
      c(Branca, Preta, Amarela, Parda, Indígena),
      ~ as.numeric(.)))

```

```

## Warning: There were 2 warnings in `mutate()`.
## The first warning was:
## i In argument: `across(c(Branca, Preta, Amarela, Parda, Indígena),
##   ~as.numeric(.))`.
## Caused by warning:
## ! NAs introduced by coercion
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.

```

Missingness

```
race_sp |>
  filter(is.na(Branca)) |>
  select(Codigo, Branca)
```

```
## # A tibble: 0 × 2
## # i 2 variables: Codigo <chr>, Branca <dbl>
```

```
race_sp |>
  filter(is.na(Amarela)) |>
  select(Codigo, Amarela)
```

```
## # A tibble: 2 × 2
##   Codigo Amarela
##   <chr>     <dbl>
## 1 3514957      NA
## 2 3555901      NA
```

```
race_sp |>
  filter(is.na(Preta)) |>
  select(Codigo, Preta)
```

```
## # A tibble: 0 × 2
## # i 2 variables: Codigo <chr>, Preta <dbl>
```

```
race_sp |>
  filter(is.na(Indígena)) |>
  select(Codigo, Indígena)
```

```
## # A tibble: 73 × 2
##   Codigo Indígena
##   <chr>     <dbl>
## 1 3500402      NA
## 2 3503000      NA
## 3 3506201      NA
## 4 3507159      NA
## 5 3507753      NA
## 6 3508900      NA
## 7 3514205      NA
## 8 3514304      NA
## 9 3515129      NA
## 10 3515806      NA
## # i 63 more rows
```

Indígena: missing for 73 municipalities

Amarela: missing for 2 municipalities

Preta: no missing

Branca: no missing

Raw Distribution

```
#create quartiles
race_sp <- race_sp %>%
  mutate(
    Branca_Q    = ntile(desc(Branca), 4),
    Preta_Q     = ntile(desc(Preta), 4),
    Amarela_Q   = ntile(desc(Amarela), 4),
    Parda_Q     = ntile(desc(Parda), 4),
    Indigena_Q  = ntile(desc(`Indígena`), 4)
  )

#table 1
race_long <- race_sp %>%
  pivot_longer(
    cols = c(Branca, Preta, Amarela, Parda, `Indígena`),
    names_to = "Race",
    values_to = "Percent"
  ) %>%
  group_by(Race) %>%
  mutate(
    Quartile = ntile(desc(Percent), 4)
  ) %>%
  group_by(Race, Quartile) %>%
  summarise(
    n = n(),
    Mean = mean(Percent, na.rm = TRUE),
    SD = sd(Percent, na.rm = TRUE),
    Min = min(Percent, na.rm = TRUE),
    Max = max(Percent, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## Warning: There were 4 warnings in `summarise()`.
## The first warning was:
## i In argument: `Min = min(Percent, na.rm = TRUE)`.
## i In group 5: `Race = "Amarela"`, `Quartile = NA`.
## Caused by warning in `min()`:
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 3 remaining warnings.
```

race_long

```
## # A tibble: 22 × 7
##   Race      Quartile      n      Mean      SD      Min      Max
##   <chr>      <int> <int>   <dbl>   <dbl> <dbl>   <dbl>
## 1 Amarela      1   161    1.52    0.993   0.76   10.3
## 2 Amarela      2   161    0.519   0.112   0.36    0.76
## 3 Amarela      3   161    0.271   0.0474  0.2     0.36
## 4 Amarela      4   160    0.124   0.0497  0.02    0.2
## 5 Amarela     NA     2 NaN      NA      Inf    -Inf
## 6 Branca       1   162   71.2    3.28   66.9    80
## 7 Branca       2   161   64.1    1.68   61.0   66.9
## 8 Branca       3   161   56.7    2.49   52.6   60.9
## 9 Branca       4   161   46.9    4.81   22.0   52.6
## 10 Indígena    1   143    0.453   1.61    0.1   15.1
## # i 12 more rows
```

Distribution Grouping Race

```
race_sp <- race_sp %>%
  mutate(
    Other = Amarela + Indígena + Preta)
```

```
race_vars <- c("Branca", "Parda", "Other")

quartile_summary <- map_dfr(race_vars, function(var) {

  race_sp %>%
    select(all_of(var)) %>%
    rename(value = all_of(var)) %>%
    mutate(
      Quartile = ntile(desc(value), 4)
    ) %>%
    group_by(Quartile) %>%
    summarise(
      n = n(),
      pct = round(100 * n() / nrow(race_sp), 1),
      median = median(value, na.rm = TRUE),
      min = min(value, na.rm = TRUE),
      max = max(value, na.rm = TRUE),
      .groups = "drop"
    ) %>%
    mutate(
      Race = var,
      Range = paste0(round(min, 1), "-", round(max, 1))
    ) %>%
    select(Race, Quartile, n, pct, median, Range)
})
```

```
## Warning: There were 2 warnings in `summarise()`.
## The first warning was:
## i In argument: `min = min(value, na.rm = TRUE)` .
## i In group 5: `Quartile = NA` .
## Caused by warning in `min()` :
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

quartile_summary

```
## # A tibble: 13 × 6
##   Race    Quartile      n    pct median Range
##   <chr>      <int> <int> <dbl> <dbl> <chr>
## 1 Branca         1   162  25.1  70.7 66.9-80
## 2 Branca         2   161   25   64.3 61-66.9
## 3 Branca         3   161   25   56.7 52.6-60.9
## 4 Branca         4   161   25   48.0 22-52.6
## 5 Parda          1   162  25.1  43.5 39.1-62.9
## 6 Parda          2   161   25   36.0 32.8-39.1
## 7 Parda          3   161   25   29.6 27.8-32.8
## 8 Parda          4   161   25   24.5 14.9-27.7
## 9 Other          1   143  22.2   9.35 7.8-19.9
##10 Other          2   143  22.2   7.06 6.3-7.8
##11 Other          3   143  22.2   5.81 5.1-6.3
##12 Other          4   142   22   4.12 2.1-5.1
##13 Other         NA    74  11.5   NA  Inf--Inf
```

Neat Table

```
quartile_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Race Quartile",
    col.names = c(
      "Race",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

Distribution of Municipalities by Race Quartile

Race	Quartile	n	%	Median (%)	Range
------	----------	---	---	------------	-------

Race	Quartile	n	%	Median (%)	Range
Branca	1	162	25.1	70.7	66.9–80
	2	161	25.0	64.3	61–66.9
	3	161	25.0	56.7	52.6–60.9
	4	161	25.0	48.0	22–52.6
Parda	1	162	25.1	43.5	39.1–62.9
	2	161	25.0	36.0	32.8–39.1
	3	161	25.0	29.6	27.8–32.8
	4	161	25.0	24.5	14.9–27.7
Other	1	143	22.2	9.4	7.8–19.9
	2	143	22.2	7.1	6.3–7.8
	3	143	22.2	5.8	5.1–6.3
	4	142	22.0	4.1	2.1–5.1
	NA	74	11.5	NA	Inf–Inf

Normality

```
race_vars <- c("Branca", "Parda", "Other")

race_sp %>%
  select(all_of(race_vars)) %>%
  pivot_longer(
    cols = everything(),
    names_to = "race",
    values_to = "value"
  ) %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ race, scales = "free") +
  theme_minimal()

## Warning: Removed 74 rows containing non-finite outside the scale range
## (`stat_bin()`).
```